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Automated AI for real-time sperm selection in ICSI: reducing variability and studying the role of sperm in embryo development

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Abstract

Background The application of Artificial Intelligence (AI) to sperm selection during Intracytoplasmic Sperm Injection (ICSI) procedures represents one of the most innovative advances in assisted reproductive technology (ART). Traditional sperm selection relies heavily on the subjective assessment of embryologists, which can lead to variability in outcomes. This study aimed to evaluate the performance of an AI-based software, Sperm ID (SiD™) v.1.0, for sperm selection during ICSI and to compare its outcomes with those obtained by experienced embryologists. Additionally, the study assessed the potential impact of sperm and oocyte quality, particularly in autologous versus donor oocyte cycles.

Methods A single-center, blind, observational study was conducted involving 102 infertile couples—60 undergoing treatment with autologous oocytes and 42 using oocytes from a donation program. Semen samples were analyzed in real time with SiD™ v.1.0, a software that quantifies progressive motility parameters and assigns each sperm a categorical score ('Best,' 'Good,' 'Medium,' or 'low'). Spermatozoa and oocytes were individually tracked from injection to embryo development. Oocyte quality was retrospectively analyzed using another AI tool, Magenta IVF R3.0. The performance of the Artificial Intelligence Sperm Selection (AISS) system was compared with that of senior embryologists (> 300 ICSI cycles/year). Statistical analysis included descriptive statistics and inferential tests to compare fertilization and embryo development rates across sperm categories and between autologous and donor cycles.

Results Biological outcomes—such as fertilization and blastocyst development—were generally similar across all sperm quality categories. However, in cycles with autologous oocytes, the use of top-quality sperm ('Best' category) was associated with a significantly higher blastocyst formation rate. In contrast, no significant differences were observed in donor oocyte cycles, regardless of sperm quality. The AISS system demonstrated comparable performance to that of senior embryologists, with similar fertilization and embryo development rates.

Conclusions The study highlights the promising role of AI-based tools in standardizing and enhancing sperm selection during ICSI. While AI-driven sperm selection showed limited impact in donor cycles, it may offer a distinct advantage in cases involving compromised oocyte quality. Furthermore, AISS may improve laboratory efficiency and support junior embryologists by reducing selection time and increasing procedural consistency.

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Keywords Automated sperm selection, Artificial Intelligence, Oocyte quality, Fertilization, Blastocyst formation, Embryo quality

Background

The application of assisted reproductive technologies (ART) has grown significantly in recent years, with nearly half a million treatments conducted annually for a population of 300 million people. This equates to an average of 1581 cycles per million inhabitants, according to the European IVF Monitoring Consortium (EIM) for the European Society of Human Reproduction and Embryology (ESHRE). Among all the procedures carried out, ICSI is the most used technique. Initially developed in 1992 to address severe male infertility [1], ICSI has now become a routine practice in clinical settings, frequently chosen even when there is no male infertility factor present [2, 3].

The human fertilization process involves a series of intricate biological events that require functionally competent sperm [4, 5]. Under natural conditions, the female reproductive tract enforces a stringent selection process for sperm [6, 7]. However, with ICSI, embryologists choose sperm based on their morphology and motility, which does not necessarily prevent the use of immature or dysfunctional sperm that appear normal [8, 9]. This bypasses the natural selection of the most optimal sperm [5], potentially leading to the selection of chromosomally abnormal sperm or those with severe molecular defects. This can negatively impact the success rates of assisted reproductive treatments, manifesting as fertilization failure, arrested embryonic development, or miscarriage [8–10].

Sperm selection is a crucial step in ICSI cycles to achieve reproductive success [2]. Once seminal samples are processed mainly through Swim-up or Density Gradient Centrifugation in the andrology laboratories, they are located at Polyvinylpyrrolidone (PVP) drops in the ICSI dishes, where embryologists have to visually analyze them and select the best sperm to microinject into the oocytes [2]. Phenotypic analysis is based on the morphology and progressive motility of the sperm. However, the current sperm selection process is not standardized, as the World Health Organization (WHO) guidelines are subjectively interpreted by clinical experts [11]. This makes the sperm selection procedure in ICSI a complex, tedious, and slow task, laden with subjectivity and intra- and inter-operator variability [12]. Some innovative techniques have emerged to improve sperm selection during ICSI procedure, such as Physiological Intracytoplasmic Sperm Injection (PICSI) [13] or Intracytoplasmic Morphologically Selected Sperm Injection (IMSI) [14]. However, the effect of these techniques on clinical outcomes

remains controversial [14–16], limiting their introduction into routine clinical practice [16].

The incorporation of AI into the sperm selection process is one of the most explored areas in recent years within the field of assisted reproduction [17]. Currently, AI is already aiding embryo selection in some fertility clinics, thanks to new algorithms capable of scoring each individual embryo, considering the embryo's morphokinetic data based on time-lapse images and videos [18, 19]. In addition, several AI tools have been developed for analyzing oocyte quality according to their morphology [20–22].

First attempts to automatize seminal samples assessment came with the introduction of the Computer-Aided Systems Analysis (CASA), which are semi-automatic and fully automatic platforms capable of objectively analysing hundreds of spermatozoa in a short period of time [23]. However, traditional CASA systems are designed to analyse large populations of spermatozoa in a single sample, which limits their ability to evaluate individual sperm characteristics in detail. Furthermore, these platforms are primarily used for diagnostic purposes and do not offer functional or selection capabilities [12]. To overcome these limitations, new tools based on AI have been developed in recent years, with the aim of enabling more accurate analysis at the individual cell level [24, 25]. Additionally, several research have combined AI models and computer vision to automate the selection of a single high-quality sperm during the ICSI procedures, as the Sperm ID (SiD) software, developed by Mendizabal-Ruiz and colleagues [26]. SiD™ (v.1.0, IVF2.0, Ltd) provides a categorical score for each individual sperm based on the evaluation of progressive motility patterns [26, 27]. It has achieved a 100% efficacy rate in parameter evaluation, classifying each sperm as 'Low', 'Medium', 'Good' or 'Best' [26]. Additionally, some studies have found a correlation between individual sperm morphology and motility patterns analyzed by SiD™, in samples ready for ICSI [28]. Finally, recent prospective studies on sibling oocytes have found comparable biological outcomes between those microinjected with SiD-selected sperm and those microinjected without SiD™ assistance [29].

However, these studies have several limitations, such as the small sample size on the study of Mendizabal-Ruiz and colleagues [26], or the seminal preparation prior ICSI using Zymot; a non-usual technique for sperm selection in andrology laboratories [29]. In addition, the study of both gametes, oocyte and sperm, simultaneously by AI tools has not been performed yet.

Therefore, the objective of this research is to provide novel insights into the correlation between SiD™ scores and reproductive outcomes across a wide range of oocyte quality populations. Moreover, the simultaneous evaluation of both sperm and oocyte quality using AI-based tools, and the analysis of each gamete's contribution to fertilization and embryo development, have not yet been reported. To our knowledge, this is the first study to integrate multiple AI tools for the assessment of not only gamete quality but also embryo quality.

Material and methods

Ethical approval

The procedure and protocol were approved by our institutional review board and ethical committee (code: 2501-VLC-037-MM) regulating and approving database analysis and clinical in vitro fertilization (IVF) procedures for research at IVIRMA Valencia. This project complies with Spanish law governing assisted reproductive technologies (14/2006).

Study design and population

This is a single-centre, non-intervention and simple blind study based on a cohort of 102 videos of ICSI procedures (60 from autologous ICSI cycles and 42 from oocyte donation program cycles) which results in a total of 628 mature oocytes inseminated (357 autologous oocytes and 271 donated oocytes). According to the recommendations of the SiD™ software developers, all semen samples were required to have a minimum concentration of 1 million spermatozoa per milliliter after sperm processing.

Fresh sperm samples obtained by masturbation were processed using swim-up or centrifugation density gradients before ICSI, and then, they were assessed in real-time with Sperm identification (SiD™) software (v.1.0) (IVF2.0 Ltd; Mexico) at ICSI dishes; which classify each individual sperm as 'Low', 'Medium', 'Good' or 'Best' according to their progressive motility parameters. Additionally, oocytes were also retrospectively assessed by an AI tool which provides a score for each oocyte (Magenta IVF R3.0; FutureFertility, Canada), related with oocyte quality [22]. In the vast majority of cases ($n=79$), half of the oocytes of each patient were included in the study, recording sperm selection and SiD™ assessment, for the subsequent analysis. If the patient had 6 oocytes MII or less, all oocytes were included. Injected sperm and their corresponding oocytes were individually followed up to assess fertilization status, blastocyst formation, embryo quality and euploid rate in those cycles with the Preimplantation Genetic Test of Aneuploidies (PGT-A) on the embryos. In all cases, sperm selection was blind to the embryologist who performed the ICSI procedure, who selected the best sperm according to their subjective

criteria and without considering the categorical score assigned by SiD™ (Fig. 1).

Inclusion and exclusion criteria

Inclusion criteria

- ICSI treatments.
- Seminal samples obtained by masturbation with a concentration equal or higher than 1 million of sperm per milliliter after sperm processing.

Exclusion criteria

- Seminal samples with a sperm concentration minor to 1 million sperm per milliliter after swim-up.
- Seminal samples of testicular biopsy.
- ICSI cycles with Calcium ionophore application.
- Patients diagnosed with recurrent pregnancy loss.

Ovarian stimulation, oocytes aspiration and AI assessment

Ovarian stimulations were performed as described by Bori et al. [19]. In brief, donors underwent ovarian stimulation using a Gonadotropine Releaser Hormone (GnRH) agonist (Decapeptyl), administered until at least eight follicles reached ≥ 18 mm in diameter, followed by transvaginal oocyte retrieval 36 h later. Recipients' endometrial preparation involved hormone replacement therapy, and post-embryo transfer, patients received 400 mg of vaginal micronized progesterone twice daily for luteal-phase support.

For ovarian stimulation, GnRH antagonist protocols were primarily used, occasionally supplemented by long GnRH agonist protocols or without LH surge suppression. Gonadotrophins, with doses ranging from 150 to 300 IU/day (Gonal-F, Puregon, Bemfola), sometimes combined with 75 IU highly purified HMG (Menopur), were started during the initial days of menstruation or after stopping contraceptive pills. Estrogens were typically given from the midluteal phase of the previous cycle for scheduling.

On stimulation Day 5, the first transvaginal ultrasound was performed, followed by monitoring every 2 days, along with serum estradiol and progesterone levels. A daily dose of 0.25 mg GnRH antagonist (Orgalutran or Cetrotide) was introduced when at least one follicle reached 14 mm. Final oocyte maturation in antagonist cycles was triggered with 0.2 mg GnRH α (Decapeptyl) and/or recombinant hCG (Ovitrelle) when three follicles reached 17–18 mm, whereas in long GnRH agonist cycles, only recombinant hCG was used.

Transvaginal ovarian puncture was carried out by follicular aspiration. Oocytes were washed with Gamete Medium (Cook Medical®, Australia) and cultured with fertilization Medium (Origio, Cooper Surgical®,

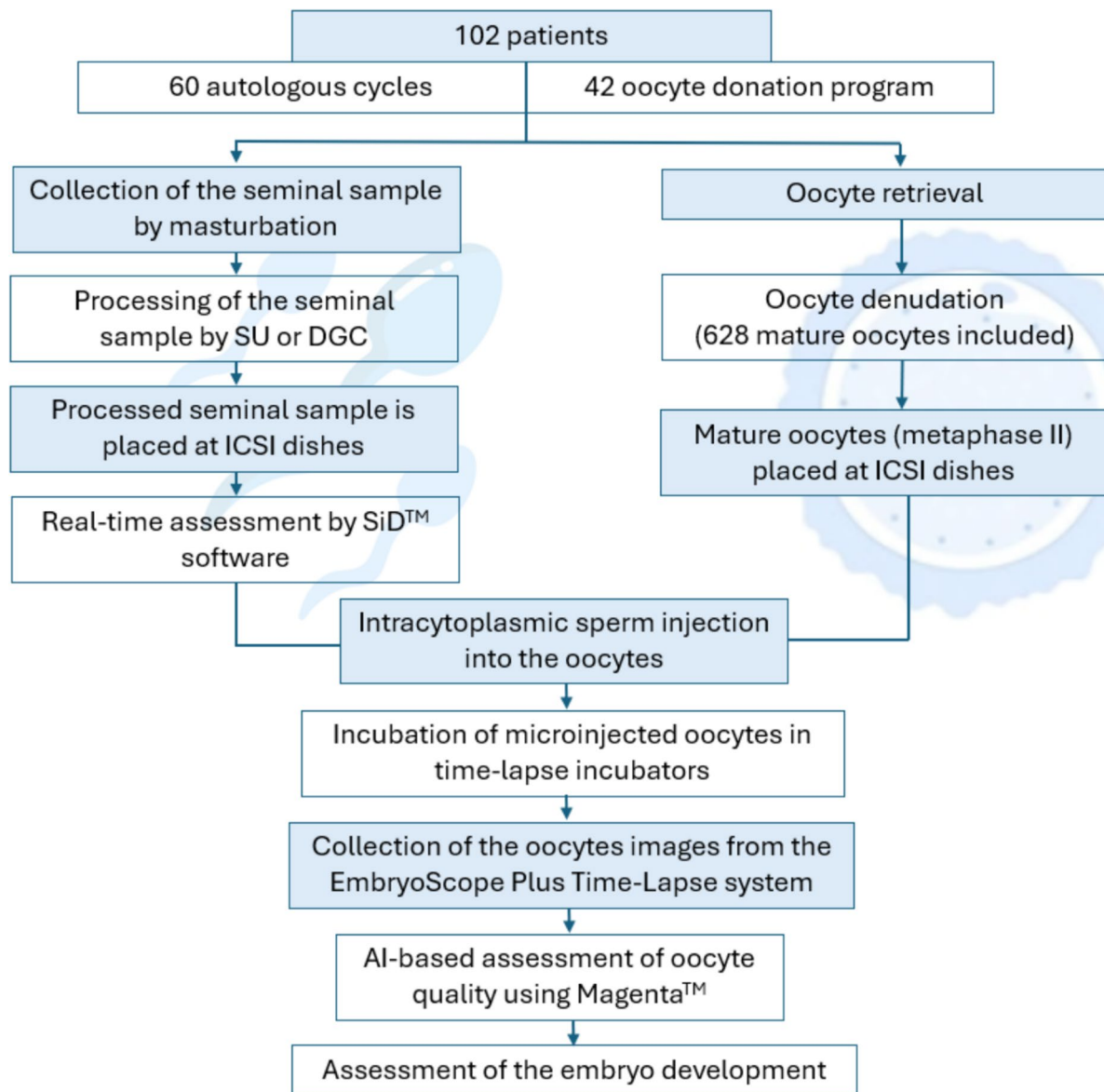


Fig. 1 Graphical abstract of the study design

Denmark) in 6% CO₂, 5% O₂ at 37 °C. Denudation was performed 4 h after oocyte retrieval by means of mechanical and chemical procedures (pipetting in 40 IU/mL of hyaluronidase).

Oocytes were retrospectively assessed by an AI tool (Magenta IVF R3.0; FutureFertility, Canada) which assigns an individual score based on the oocyte morphology. Images of oocytes were taken from the time-lapse incubators 0 h after microinjection. The oocytes' evaluation post-ICSI by this AI tool have been tested and revealed a better predictive accuracy for fertilization than pre-ICSI assessment, and similar for the blastocyst formation [30].

Semen processing, SiD™ analysis and ICSI

Seminal samples were processed by Swim-up or Density Gradients as described by Meseguer et al., 2024. Then, seminal samples were located at PVP drops (10%) at ICSI plates to perform the ICSI procedure at Olympus IX7 microscope. The selection, immobilization and microinjection of the sperm was recorded using a digital camera (Olympus U-Tv0.5XC-3) connected to an inverted optical microscope (Olympus IX7). Quantitative analyses of motility parameters were performed by SiD™ (v.1.0, IVF 2.0, LTD), which can detect a maximum of 3 sperm simultaneously, computing for each one of them the values of LIN (*Linearity*), VSL (*Straight-Line Velocity*) and the HMP (*Head Movement Pattern*) parameters. With them, SiD™ generated a categorical score ('Best', 'Good',

'Medium' or 'Low'), operating at 15 frames per second, with a resolution of 640×480 pixels (Fig. 2). The software criteria for the initial detection of the sperm are unknown for the investigators of this study.

Embryo incubation

Microinjected oocytes were incubated in Embryoscope™ (ES) (Vitrolife, Denmark) and Geri® (Genea Biomedx, Australia) time-lapse incubators with gassed culture Medium and stable conditions throughout embryonic development: 37 °C, 6% CO₂, and 5% O₂. The development of each sperm-oocyte pair was monitored.

Fertilization was confirmed 16–18 h after microinjection by the presence of two pronuclei (PN) and two polar bodies (PB). The embryos were kept in the incubator until day 5 (D5) or day 6 (D6) of development, depending on the optimal transfer date.

Embryo evaluation

Embryonic development was evaluated up to D5 or D6 (in some cases), corresponding to the blastocyst stage. The embryos were classified according to the 2021 criteria of the Association for the Study of Reproductive Biology (ASEBIR), based on their morphology, assessing

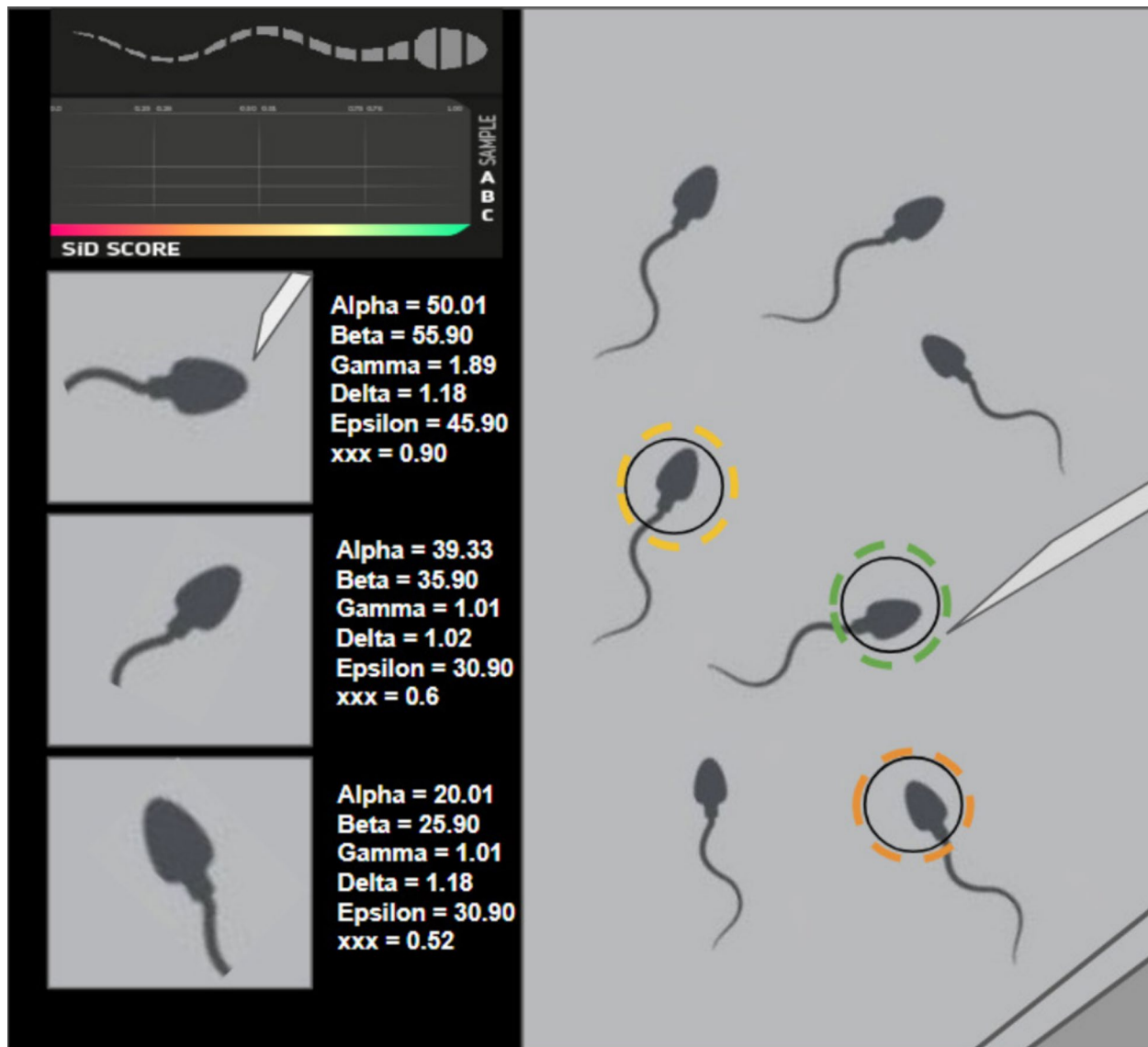


Fig. 2 User's graphical interface of SiD™ software (v.1.0, IVF2.0 Ltd; Mexico). Simultaneous analysis of three individual spermatozoa is depicted, with a distinct color circle assigned to each. The same spermatozoa are represented on the left panel, ranked according to a gradient from optimal to suboptimal performance. Parameters labeled as 'Alpha', 'Beta', 'Gamma', 'Epsilon', etc., correspond to sperm trajectory and motility metrics assessed by the SiD™ software. These parameters, proprietary to the company, remain undisclosed to the investigators. Green sperm is categorized as 'Best', yellow sperm is categorized as 'Good' and orange sperm is categorized as 'Medium' and red sperm is categorized as 'Low' (red sperm is not represented because SiD™ system only assessed 3 sperm simultaneously)

the inner cell mass (ICM) and the trophectoderm (TE) (Supplementary tables 1,2,3 and 4). In addition, three different AI algorithms were applied to assess embryo quality of each embryo located at EmbryoScope Plus incubators (IDAScore v.2.0., KIDScore D5 v.3.1; HERA v.1.0.). Embryos maintained in culture at EmbryoScope incubators were assessed by KIDScore D5 v.3.1 and HERA v.1, and those at GERI incubators were retrospectively assessed only by HERA v.1.0 IDAScore v.2.0 and KIDScore D5 v.3.1 ranged embryo with a score from 1 (low likelihood of implantation) to 9.9 (high likelihood of implantation) and they get the images and videos of the embryo development automatically from the EmbryoScope. To get scores from HERA v.1.0 (Alife), we uploaded the images. Scores varied from 0 to 1. Based on these algorithms, the best-quality embryo was selected for transfer (fresh cycle), and the remaining embryos were vitrified for a possible subsequent deferred cycle. Finally, the definition of usable blastocysts differed according to the type of cycle. In PGT-A cycles, usable blastocysts were defined as those that underwent biopsy for chromosomal testing, out of the total number of blastocysts obtained. In cycles without genetic testing, usable blastocysts were considered to be those either transferred in a fresh cycle or vitrified for deferred transfer.

PGT-A

Embryos enrolled in the PGT-A program ($n=111$) were taken out of the incubator on the third day of development. A small laser cut was made through the zona pellucida (assisted hatching) using the Hamilton-Thorne

Lykos® laser. On the fifth day of culture, a biopsy was conducted by extracting about five trophectoderm cells with a biopsy pipette. Chromosome analysis was then performed utilizing next generation sequencing technology (Thermo Fisher Scientific, Waltham, MA, USA) [30].

Statistical analysis

First, a descriptive analysis of the data was performed, where continuous values were studied according to the mean, standard deviation and confidence intervals. Then, normality and variance homogeneity were studied using the Shapiro Wilk Test and Levene, respectively. To assess statistical differences between the outcome variables, hypothesis testing was performed, including ANOVA for numerical variables, Chi-square Test for the categorical ones and post-hoc tests as Tukey or Bonferroni. All analyses were performed using IBM SPSS statistics software (New York, USA). $P<0.05$ was considered significant for all tests.

Results

A total of 102 couples were included in the present study. Of these, 42 belonged to the oocyte donation program and 60 couples used their autologous oocytes. The characteristics of the women included in the study are shown in Table 1A. The majority were over 38 years old and with a body mass index (BMI) and ovarian reserve within the normal range. Characteristics of the seminal samples are included in Table 1B.

First, we studied biological outcomes on the total population. The fertilization rate achieved by each

Table 1 Characteristics of the total study population, patients using their autologous oocytes and patients from the oocyte donation program

	Total study Population	AUTOLOGOUS OOCYTE PATIENTS	OOCYTE DONATION PATIENTS
A			
Patients age (years)	39.48 ± 4.69	37.57 ± 4.15	42.32 ± 3.99
Donors age (years)	-	-	25.49 ± 4.98
BMI (kg/m ²)	23.60 ± 3.83	23.706 ± 3.86	23.46 ± 3.83
Days to transfer (% patients)	88.24%(D5)/11.76%(D6)	83.87%(D5)/16.13%(D6)	90.48%(D5)/9.52%(D6)
Number of retrieved/donated oocytes	12.31 ± 7.14	12.72 ± 8.01	11.10 ± 3.31
Number of MII oocytes microinjected	9.89 ± 5.01	8.98 ± 5.96	11.24 ± 2.63
Number of fertilized oocytes (2PN + 2 PB)	7.93 ± 4.18	7.11 ± 4.76	9.15 ± 2.74
Number of MII oocytes recorded *	6.82 ± 3.91	7.00 ± 4.47	6.56 ± 2.90
B			
Partner age	41.35 ± 5.98	39.70 ± 5.49	43.95 ± 5.85
Volume fresh sample (FS)	2.29 ± 1.23	2.53 ± 1.24	1.93 ± 1.148
Volume selected sample (SS)	0.44 ± 0.09	0.44 ± 0.99	0.43 ± 0.076
Sperm Concentration (mill/mL) FS	48.29 ± 30.83	49.36 ± 31.29	46.69 ± 30.47
Sperm Concentration (mill/mL) SS	6.94 ± 7.00	7.54 ± 7.76	6.02 ± 5.63
Total sperm FS	111.22 ± 90.33	127.12 ± 99.56	87.56 ± 69.11
Total sperm SS	2.99 ± 2.95	3.24 ± 3.24	2.60 ± 2.44

Results are presented with the mean and plus/less Standard deviation (SD); except from Days transfer with the percentage of patients with D5 transfer or D6 transfer

*Number of MII oocytes recorded corresponds to those included in the study and assessed for sperm selection using SiD™

Table 2 Biological outcomes in the total population, patients using autologous oocytes and patients using donated oocytes. Table 2A includes results from fertilization to blastocyst formation and euploid blastocyst rate. Table 2B includes results about embryo quality (based on ASEBIR categories and AI-algorithms D5). *ND* = non-data available

Based on ASEBIR categories and AI algorithms D5; ND – non data available					
A)	Fertilization rate (%, IC95%)	Blastocyst formation rate (%, IC95%)	Usable blastocyst rate (%, IC95%)	Euploid blastocyst rate (%, IC95%)	
Best	79.89 (74.04–85.74)	70.07 (62.58–77.66)	44.02 (36.78–51.26)	55.00 (38.89–71.11)	
Good	81.38 (76.49–86.26)	68.16 (61.66–74.65)	44.94 (38.69–51.18)	47.62 (31.87–63.37)	
Medium	78.20 (71.09–85.30)	65.38 (56.09–74.69)	40.60 (32.15–49.06)	30.43 (10.09–50.78)	
Low	84.38 (75.23–93.51)	51.85 (38.09–65.62)	35.94 (23.86–48.01)	50.00 (0–107.47)	
B)	ASEBIR CRITERIA		AI-algorithms D5		
	Top-rank embryo (%, IC95%)	Non-viable embryo (%, IC95%)	KIDScore (mean, IC95%)	IDAScore (mean, IC95%)	HERA (mean, IC95%)
Best	59.22 (49.57–68.87)	17.48 (10.01–24.93)	5.69 (5.05- 6.33)	4.85 (3.79–5.90)	0.312 (0.275–0.349)
Good	62.77 (54.58–70.97)	17.52 (11.07–23.96)	5.59 (5.01–6.17)	5.18 (4.41–5.96)	0.328 (0.295–0.362)
Medium	60.29 (48.36–72.23)	22.06 (11.94—32.17)	4.70 (3.95- 5.46)	3.99 (2.89—5.11)	0.315 (0.269 –0.361)
Low	60.71 (41.43–80.00)	17.86 (2.73—32.98)	5.32 (3.97–6.67)	5.16 (3.11–7.21)	0.318 (0.252–0.383)

sperm quality category was very similar between them (Table 2A). Related to the blastocyst rate our results suggest a non-significant increase in both of them, as sperm quality increases [51.85% ('Low'); 65.38% ('Medium'); 68.16% ('Good'); 70.07% ('Best')]. Related to the usable blastocyst, sperm categorized as 'Best' or 'Good' achieved the highest percentages (44.02% and 44.94% respectively), being lower for the 'Medium' and 'Low' sperm categories (40.60% and 35.94% respectively) (Fig. 1). Overall, cumulative outcomes were more favorable in the 'Best' and 'Good' sperm quality categories compared with 'Medium' and 'Low' (Fig. 3).

Related to the embryo quality, non-differences were found in the percentage of type A, B, C or D embryos, among each sperm quality (Supplementary Table 5); neither on the percentage of Top-quality embryos (A + B) and Non-viable ones (D) (Table 2B). Additionally, we studied embryo quality on day 5 based on AI-algorithms' scores: KIDScore D5 v3.2; IDAScore v2.0 and HERA v1.0. The assessment by these AI tools found that there were no differences in the mean score assigned to the embryos of each type of sperm-quality group (Table 2B). Finally, euploid blastocyst rate we also assessed, without observing any significant difference in the proportion of

**Fig. 3** Stacked column chart of the outcomes of the total population. Cumulative outcomes (fertilization, blastocyst formation, and usable blastocyst rates) across sperm quality categories in the total population are represented

Table 3 Biological outcomes for patients using autologous oocytes. Table 3 A includes results from fertilization to embryo blastocyst formation and euploid blastocyst rate. Table 3B includes results about embryo quality (based on ASEBIR categories and AI-algorithms D5). *ND*=non-data available

C57;ND = Non-data available					
A)	Fertilization rate (% IC95%)	Blastocyst formation rate (% IC95%)	Usable blastocyst rate (% IC95%)	Euploid blastocyst rate (% IC95%)	
Best	82.18 (74.59—89.77)	67.47 (57.18—77.16)	44.55 (34.69—54.41)	45.45 (27.52—63.38)	
Good	79.43 (72.67- 86.19)	64.29 (55.2—73.30)	41.13 (32.91—49.36)	44.74 (28.17—61.30)	
Medium	71.43 (60.58—82.29)	56.00 (41.75—70.25)	34.29 (22.89—45.69)	26.32 (4.51—48.12)	
Low	80.00 (67.85—92.15)	41.67 (24.75—58.58)	24.44 (11.38—37.50)	50.00 (7.47—107.48)	
B)	ASEBIR CRITERIA		AI-algorithms D5		
	Top-rank embryo (%, IC95%)	Non-viable embryo (%, IC95%)	KIDScore v3.2 (mean, IC95%)	IDAScore v2.0 (mean, IC95%)	HERA v1.0 (mean, IC95%)
Best	53.60 (40.09—67.05)	17.90 (7.51—28.21)	5.34 (4.46—6.23)	4.54 (3.13—5.94)	0.255 (0.21—0.30)
Good	51.30 (39.56—63.21)	19.50 (10.07—28.21)	5.28 (4.40—6.16)	5.43 (4.44—6.42)	0.257 (0.220—0.299)
Medium	53.50 (33.88—73.26)	17.90 (2.73—32.98)	4.83 (3.62—6.04)	4.81 (3.04—6.57)	0.272 (0.204—0.339)
Low	46.70 (18.07—75.26)	26.70 (1.31—52.01)	4.39 (2.27—6.50)	4.47 (1.75—7.19)	0.261 (0.174—0.349)

chromosomally normal blastocysts of each sperm-quality group (Table 2A).

The next step was studying the biological outcomes differentiating between those patients who used their autologous oocytes ($n=60$; 357 MII oocytes) and those who belonged to the oocyte donation program ($n=42$; 271 MII oocytes) (Tables 3 and 4). Results on patients who used their own oocytes showed that rates of fertilization, blastocyst formation and usable blastocyst were similar to those obtained on the total population. However, in this case, blastocyst formation rate was statistically significant higher in the ‘Best’ group compared to the ‘Low’ group (67.47% vs 41.67%; $p<0.05$) (Fig. 2). Similarly, we found that cumulative outcomes were more favorable in the ‘Best’ and ‘Good’ sperm quality categories compared with ‘Medium’ and ‘Low’ (Fig. 4).

Related to the embryo quality, our findings showed any significant difference between groups on the percentage of embryos of type A, B, C or D (Supplementary Table 6); with an increase on the percentage of Top-quality embryos (A + B) and a decrease on the non-viable embryos (type D) rate of the ‘Best’ group versus de ‘Low’ group (53.57% vs 46.67%; 17.87% vs 26.67%, $p>0.05$). However, these differences were not statistically significant. Additionally, the embryo quality assessment by KIDscore D5 v3.1, IDAScore v2.0 and Hera v1.0 did not show any significant difference between groups (Table 2B); as euploid blastocyst rate (45.45% (‘Best’); 44.74% (‘Good’) vs 26.32% (‘Medium’); 50.00% (‘Low’); $p>0.05$).

When we studied results on oocyte donation patients, we found practically identical results in all the biological outcomes assessed (Fig. 5). In contrast to the outcomes obtained in the total population and those obtained specifically in autologous oocytes cycles, cumulative outcomes were similar between the sperm categories, being even higher in the ‘Low’ group.

Additionally, any trend was found when studying embryo quality according to ASEBIR criteria (Supplementary Table 7). Results obtained by AI-algorithms were similar to those recently described (Table 4B).

Going further, oocyte quality was assessed using an AI-based tool (Magenta IVF R3.0, Future Fertility; Canada), with the aim of evaluating the individual contribution of each gamete to the processes of fertilization and embryonic development. Initially, oocytes were categorized into two groups based on their AI-derived quality score: the first group comprised oocytes with a score below 5, while the second included those with a score equal to or greater than 5. Subsequently, oocytes were also stratified according to the motility classification of the spermatozoon used for microinjection. Specifically, oocytes were grouped depending on whether they had been injected with spermatozoa of high motility (‘Best’ or ‘Good’, as defined by the SiD™ criteria) or with spermatozoa of lower motility (‘Medium’ or ‘Low’, according to the same classification system).

Although no statistically significant differences were observed among the groups, a trend toward improved outcomes was identified when poor-quality oocytes

Table 4 Biological outcomes for patients using autologous oocytes. Table 4 A includes results since fertilization to embryo blastulation and euploidy rate. Table 4B includes results about embryo quality (based on ASEBIR categories and AI-algorithms D5). *ND*= non-data available

A)	Fertilization rate (% IC95%)	Blastulation rate (% IC95%)	Usable blastocyst rate (%, IC95%)	Euploidy rate (% IC95%)	
Best	77.11 (67.78—86.34)	73.44 (62.31—84.56)	56.25 (43.76—68.74)	100.00 (100.00—100.00)	
Good	83.86 (76.86—91.06)	73.03 (63.63—82.44)	59.55 (49.15—69.94)	75.00 (0–154.56)	
Medium	85.71 (76.83—84.60)	74.07 (62.00—86.15)	55.56 (41.86—69.25)	50.00 (0–141.86)	
Low	94.74 (83.68—105.79)	72.22 (49.30—95.14)	66.67 (42.54—90.79)	ND	
B)	ASEBIR CRITERIA		AI-algorithms D5		
	Top-rank embryo (%, IC95%)	Non-viable embryo (%, IC95%)	KIDScore v3.2 (mean, IC95%)	IDAScore v2.0 (mean, IC95%)	HERA v1.0 (mean, IC95%)
Best	65.96 (51.80—80.02)	17.00 (5.86—28.18)	6.04 (5.07—7.01)	5.28 (3.51- 7.03)	0.377 (0.320—0.434)
Good	75.40 (64.63—86.14)	15.40 (6.37—24.40)	5.82 (5.03—6.61)	4.93 (3.69—6.17)	0.397 (0.349—0.446)
Medium	65.00 (49.55—80.45)	25.00 (10.98—39.02)	4.64 (3.62—5.66)	3.41 (1.90–4.92)	0.348 (0.284—0.411)
Low	76.90 (50.42—103.42)	7.70 (0—24.45)	6.57 (5.07–8.07)	6.77 (0.75—12.78)	0.388 (0.290—0.486)

(score < 5) were microinjected with Good-motility sperm (GM sperm), compared to those injected with poor-motility sperm (PM sperm). Specifically, fertilization and

blastocyst formation rates, as well as overall embryo quality were higher in the GM sperm group. Notably, a non-significant increase of approximately 10% was observed

Outcomes of autologous oocytes cycles

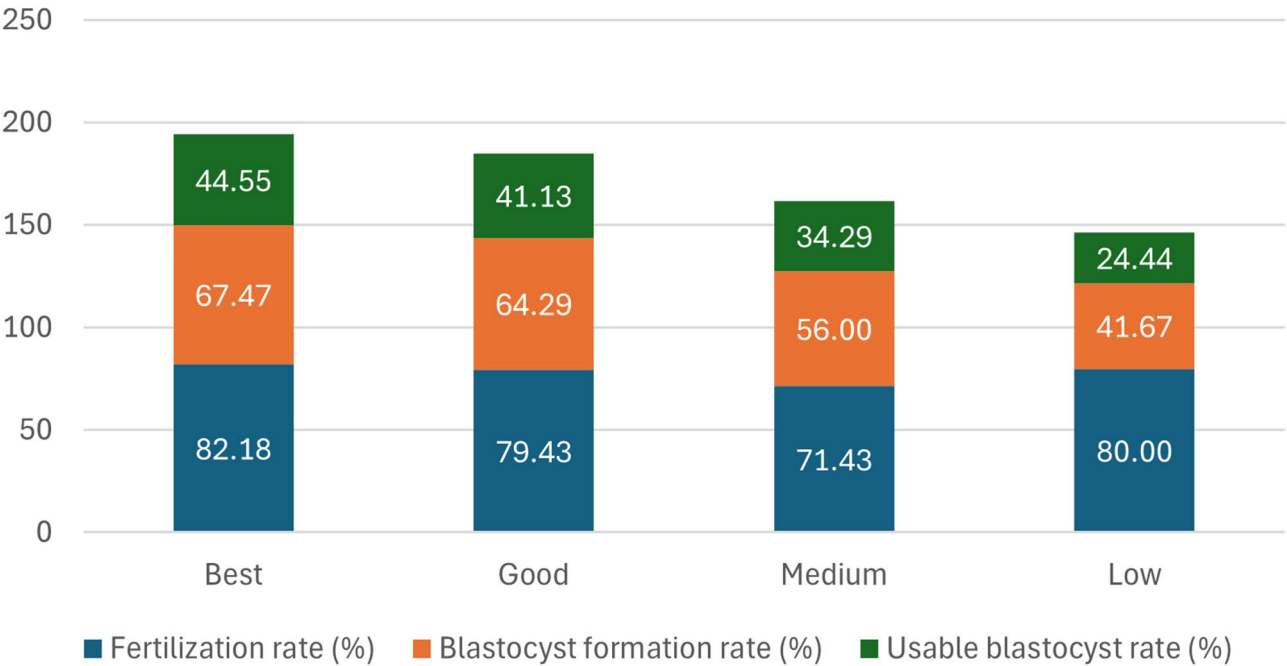


Fig. 4 Stacked column chart of the outcomes of autologous oocytes cycles. Cumulative outcomes (fertilization, blastocyst formation, and usable blastocyst rates) across sperm quality categories in the autologous oocytes cycles

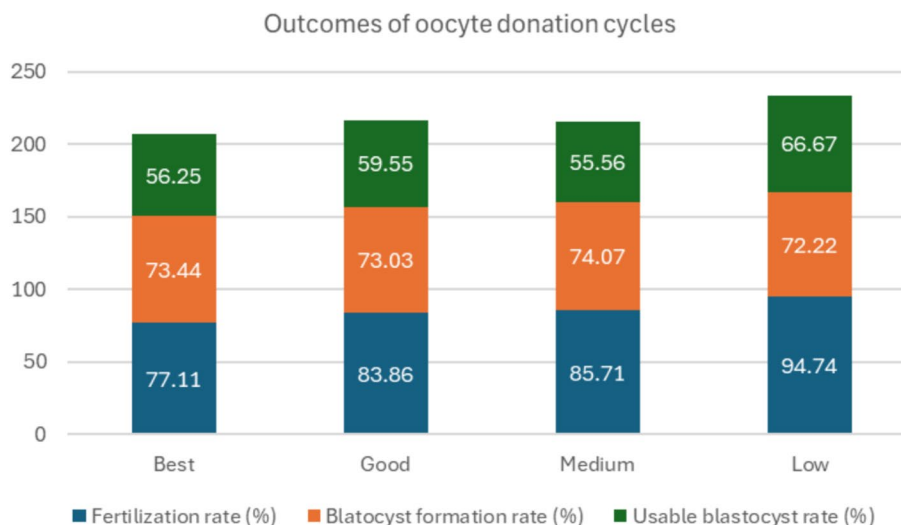


Fig. 5 Stacked column chart of the outcomes of oocyte donation cycles. Cumulative outcomes (fertilization, blastocyst formation, and usable blastocyst rates) across sperm quality categories in the oocyte donation cycles are represented

in both the usable blastocyst rate and the proportion of top-quality embryos. Therefore, the cumulative outcomes in the population of oocytes with an AI-derived score under 5 (poor-oocyte quality) and microinjected with GM sperm were more advantageous than cumulative outcomes in oocytes with poor quality and microinjected with PM sperm (Fig. 4). Regarding chromosomal status, the proportion of euploid blastocysts derived from the injection of GM sperm was twice as high as that observed in the PM sperm group (40.00% vs. 20.00%; $p > 0.05$). However, it is important to note that the number of blastocysts biopsied in each group was limited ($n = 15$ and $n = 5$, respectively).

In contrast, when analyzing the subgroup of oocytes with a quality score equal to or greater than 5 (good-oocyte quality), no meaningful differences were found between the GM and PM sperm groups in terms of fertilization rate, blastocyst development, or embryo quality (Table 3). In this case, cumulative outcomes were very similar between the group of oocytes with AI-derived score under 5 and microinjected with GM sperm and PM sperm (Fig. 6). Regarding chromosomal status, blastocysts derived from the GM sperm group exhibited a higher euploidy rate (53.85%) compared to those from the PM sperm group (37.50%), although this difference did not reach statistical significance ($p > 0.05$). Again, the small number of euploid blastocysts analyzed may limit the robustness of this comparison.

When embryo quality was assessed using AI-based scores from different evaluation tools, no statistically significant differences were observed. However, a slight trend towards higher scores was noted in embryos derived from the microinjection of GM sperm compared

with those obtained from PM sperm, both in oocytes with MAG scores below and above 5 (Table 5).

These results are in line with those mentioned, where patients who use their own oocytes, had better outcomes when 'Best' or 'Good' sperm were microinjected. Finally, we also found statistically significant differences on the oocytes' scores of patients who used their own oocytes (5.686) and those from the egg donation program (6.249) ($p = 0.012$). That correlates with the results obtained when studying biological outcomes differentiating between autologous and donated oocytes.

Finally, we performed an additional retrospective analysis to assess SiD's ability of performing sperm selection and to compare it to the sperm selection performed by senior embryologists. We studied cycles where half of the oocytes had been microinjected with sperm recorded and assessed by SiD™ software ($n = 88$ cycles). We compared biological outcomes of those oocytes microinjected with 'Best' or 'Good' sperm; which are the Good ones to microinject according to SiD™ software's criteria (assisted-ICSI group; $n = 328$ MII oocytes) and those biological outcomes of the oocytes microinjected without SiD™ assessment ($n = 357$ MII oocytes); where sperm were selected according to the embryologist criteria and without knowing categorical score assigned by SiD™.

We found a non-significant trend towards better outcomes in the assisted-ICSI group for biological outcomes including fertilization rate (FR, 79.88%), blastocyst rate (BR, 72.14%), usable-blastocyst rate (UBR, 58.40%) and euploid-blastocyst rate (EBR, 57.14%) than in the conventional ICSI group (FR = 74.79%; BR = 67.41%; UBR = 55.81%; ER = 49.06%). When studying these laboratory indicators at autologous oocytes' cycles ($n = 44$ cycles, 328 MII oocytes), differences were higher between

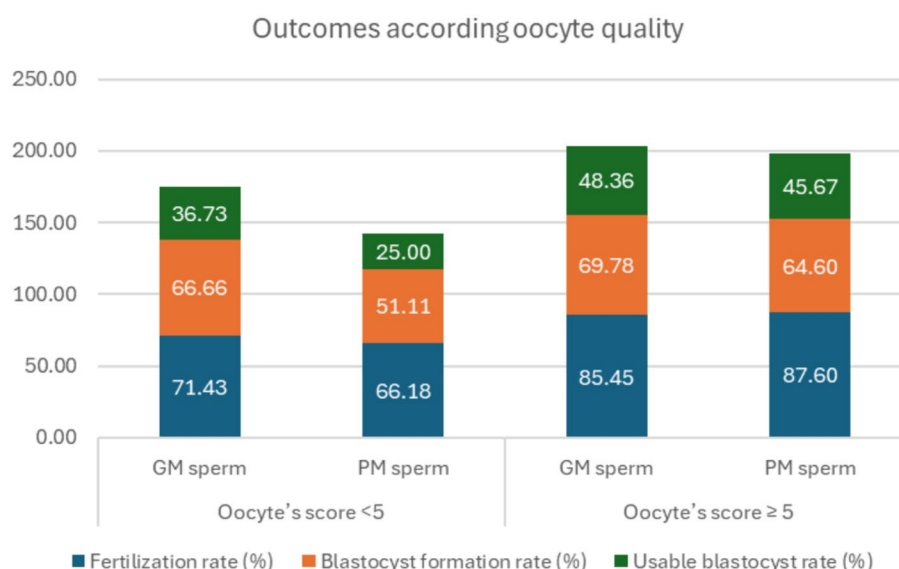


Fig. 6 Stacked column chart of the outcomes according oocyte quality assessed by AI. Cumulative outcomes (fertilization, blastocyst formation, and usable blastocyst rates) across Good-motility (GM) sperm and Poor-motility (PM) sperm group are represented

both groups, especially for blastocyst rate (70.00% vs 58.91%), usable-blastocyst rate (60.00% vs 50.39%) and euploid-blastocyst rate (48.21% vs 37.50%) ($p > 0.05$) (Table 6). Related to embryo quality, the percentage of top-rank embryo and non-viable embryo was very similar; both in the total population (61.37% vs 61.11%; 16.40% vs 17.22%) and in own oocytes' cycles (56.12% vs 53.94%; 14.26% vs 15.79%) ($p > 0.05$) (Table 6). Results in

patients of the egg donation program ($n = 36$ cycles, 357 MII oocytes) were very similar in both groups; except from euploid-blastocyst rate which was near than 30% higher in the assisted-ICSI group (Table 6). Nevertheless, the limited number of biopsied embryos could explain this result (70 biopsied embryos in A-ICSI group and 53 biopsied embryos in ICSI group).

Table 5 Biological outcomes in groups of oocytes with bad quality (score < 5) and good quality (score ≥ 5). Table 2 A includes results since fertilization to blastocyst formation and euploidy rate. Table 2B includes results about embryo quality (based on ASEBIR categories and AI-algorithms D5). All results were non-statistically significant ($p > 0.05$)

A)		Fertilization rate (% IC95%)	Blastocyst forma- tion rate (%, IC95%)	Usable blastocyst rate (% IC95%)	Euploidy rate (%, IC95%)	
Oocyte's score < 5	GM sperm	71.43 (64.04—72.82)	66.66 (57.50—75.83)	36.73 (28.85—44.62)	40.00 (11.92—68.08)	
	PM sperm	66.18 (54.64—77.71)	51.11 (35.92—66.30)	25.00 (14.44—35.46)	20.00 (0—75.53)	
Oocyte's score ≥ 5	GM sperm	85.45 (81.26—89.645)	69.78 (63.87—75.70)	48.36 (42.42—54.31)	53.97 (41.31—66.62)	
	PM sperm	87.60 (81.83—93.36)	64.60 (55.65—73.55)	45.67 (37.79—55.23)	37.50 (16.62—58.38)	
B)		ASEBIR CRITERIA		AI-Algorithms D5		
		Top-quality embryo (%)	Non- viable embryo (%)	KIDScore v3.2 (mean,IC95%)	IDAScore v2.0 (mean,IC95%)	HERA v1.0 (mean,IC95%)
Oocyte's score < 5	GM sperm	57.14 (45.26—69.03)	14.29 (5.88—22.69)	5.45 (4.90—6.01)	4.55 (3.92—5.19)	0.310 (0—0.80)
	PM sperm	47.83 (25.74—69.91)	25.09 (6.67—45.50)	4.34 (3.38—5.30)	3.00 (1.98—4.03)	0.280 (0—1.15)
Oocyte's score ≥ 5	GM sperm	63.41 (55.97—70.86)	19.51 (13.38—25.64)	5.74 (5.30—6.18)	5.37 (4.87—5.86)	0.327 (0—0.67)
	PM sperm	64.38 (53.14—75.63)	19.18 (9.93—28.43)	5.13 (4.46—5.80)	5.01 (4.24—5.79)	0.327 (0—0.83)

Table 6 Biological outcomes between ICSI group and A-ICSI (assisted-ICSI) group in the total population; cycles using autologous oocytes and cycles using donated oocytes

	Total population		Autologous oocytes		Donated oocytes	
	ICSI	A-ICSI	ICSI	A-ICSI	ICSI	A-ICSI
Fertilization rate (%)	74.79	79.88	79.54	78.18	78.34	72.08
Blastocyst formation rate (%)	67.41	72.14	58.91	70.00	73.98	73.94
Usable blastocyst rate (%)	55.81	58.40	50.39	60.00	56.09	58.86
Top-rank embryo rate (%)	61.11	61.37	53.94	56.12	67.03	66.66
Non-viable embryo rate (%)	16.40	17.22	14.26	15.79	18.68	18.08
Euploidy-blastocyst rate (%)	49.06	57.14	37.50	48.21	66.67	92.86

Discussion

Quantitative analysis of sperm motility patterns by AI tools is primarily based on recommendations from the World Health Organization (WHO), such as Straight-Line Velocity (VSL), Linearity (LIN), Average Path Velocity (VAP), Curvilinear Velocity (VCL), Amplitude of Lateral Head Displacement (ALH) or Straightness (STR) [31]. These motility parameters have become more reliable indicators of sperm functionality compared to conventional semen profile parameters [32].

This preliminary study involves the first version of an Artificial Intelligence Sperm Selection (AISS) software which generates a score to each individual sperm ('Low', 'Medium', 'Good' or 'Best') based on progressive sperm motility parameters: VSL and LIN, related to the sperm trajectory. It also assesses the head movement pattern (HMP) which have been associated with the flagellum's helicoid beat pattern and it is related to the quality of spermatozoa [26, 33]. The features of each spermatozoon are processed and then evaluated using a mathematical model that determines the quality of each spermatozoon and ranks them accordingly. The result of the ranking is shown to the user in real-time [26].

Results in our total population were very similar across all sperm categories assigned by SiD™ software. A numerical tendency toward higher blastocyst and usable blastocyst rates was observed when microinjection was performed with sperm classified as 'Best' or 'Good' compared to 'Medium' or 'Low'. Nevertheless, these differences did not reach statistical significance. Therefore, no firm conclusions can be drawn, and the observed patterns should be regarded only as exploratory. When studying biological outcomes on patients using their autologous oocytes, statistically significant differences were observed. Particularly, blastocyst formation rate was significantly higher in those oocytes microinjected with 'Best' sperm compared to those microinjected with sperm categorized as "Low". These results are in line with those previously published by Mendizabal and colleagues (2022), who found that higher SiD™ scores correlate with higher blastocyst formation rates. Those authors also identified a correlation between SiD™ scores and fertilization rate, showing that good quality sperm (those

classified as 'Best' or 'Good') achieved statistically significant higher fertilization rates than sperm with poor motility patterns. However, sample size in that pioneering study was limited, with 383 sperm assessed by SiD™ and only 78 sperm microinjected. Our study did not reproduce those statistically significant differences in most analysis, although it was conducted with a larger sample size, providing a broader dataset for analysis. This reinforces the relevance of the initial findings while suggesting that further studies with extensive populations are needed to fully elucidate the relationship between sperm quality categories and reproductive outcomes as fertilization and blastocyst formation rates.

Regarding embryo quality and euploidy rate, no clear or consistent trends were observed in the total population or in the oocyte donation program subgroup. In the subgroup of patients using autologous oocytes, we found a higher proportion of good-quality embryos (according to ASEBIR criteria), fewer non-viable embryos, and a higher euploidy rate in oocytes microinjected with 'Good' or 'Best' sperm; according to the AI-tool assessment. Nonetheless, these differences did not reach statistical significance, which means that the patterns observed in embryo quality and euploid blastocyst rate may be regarded as biologically suggestive, warranting confirmation in larger and properly powered studies. These exploratory results highlight the need to consider both gametes in the design of assisted reproduction strategies, while avoiding firm conclusions until stronger evidence becomes available.

Finally, the retrospective comparison between the automated sperm selection using SiD™ and this performed by senior embryologists showed that using an AISS software during ICSI procedures achieved comparable biological outcomes than the selection performed by senior embryologists. Similar conclusions were obtained in the prospective study of Montjean and colleagues (2024); which compared the results in sibling oocytes' cohorts [29]. They showed that those oocytes microinjected with SiD™ assistance selecting 'Good' or 'Best' sperm (n = 326 MII oocytes), achieved higher proportion of Good quality embryo and Top-quality embryo on day 5, than those microinjected without SiD™ assistance (n = 320 MII oocytes) [29]. Euploid-blastocyst rate was also higher in

the assisted-ICSI group. Nevertheless, differences found by those authors were non-significant and the sample size of biopsied embryos were very limited (15 embryos in the A-ICSI and 9 embryos in the ICSI group); similarly to our study. These studies suggest that AISS tools such as SiD™ significantly enhance the ICSI process by eliminating user bias, reducing fatigue, and improving time management, particularly in cases where staff may lack expertise in sperm selection. Importantly, these benefits are suggested mainly because, even if the use of SiD™ does not improve laboratory outcomes, it at least maintains them at comparable levels while optimizing the ICSI workflow and reducing operator effort.

One innovative aspect of our study is the assessment of embryo quality by several AI tools such as IDAScore v2.0, KIDScore D5 v3.1 and HERA v1.0. In addition, the assessment of the oocyte by another AI tool helped us to study the contribution of each gamete to fertilization and embryo development. Including not only autologous cycles but also oocyte donation cycles allowed for the assessment of the impact of sperm quality across different oocyte sources, providing a broader understanding of how sperm characteristics influence fertilization and embryo development in varying oocyte backgrounds. Across our analyses, all observed tendencies toward improved outcomes after the microinjection of good quality sperm (based on SiD™ assessment) were mainly seen when studying poor-quality oocytes in isolation—both in the autologous oocyte group (typically from patients of advanced maternal age) compared with donor oocytes, and in cycles involving oocytes with an AI score below 5, which also reflects reduced quality. A plausible explanation is the reparative or compensatory potential of young, good-quality oocytes, which may counterbalance sperm-related deficiencies, thereby masking the influence of sperm quality in those settings [30, 34, 35]. However, most of these differences found in this study did not reach statistical significance and should therefore be interpreted with caution. Finally, evaluating the effectiveness of the AISS software and comparing it to the sperm selection performed by senior embryologists offers a new insight on workflow in IVF laboratories. In these settings, where numerous complex tasks are performed, automating certain processes could prove highly beneficial.

The main limitation of this study lies in the proprietary and opaque nature of the algorithm underlying the AI-based sperm selection tool. Since the criteria employed by the software are not disclosed to investigators, reproducibility of the results cannot be independently verified, and this lack of transparency poses a significant challenge to scientific rigor and the broader validation of the findings. Additional limitations include a moderate sample size (102 couples) and its single-center, retrospective

design, which may restrict statistical power and the generalizability of the results. Finally, sperm gross morphology may also influence reproductive outcomes; but the first version of SiD™ used in this study considers only motility parameters. Nevertheless, future versions will allow morphologic analysis to be carried out by machine learning approaches in real time [25, 26]. Another limitation to consider is the possible inclusion of samples with low sperm quality, since the threshold was set at 1 million sperm per milliliter, after sperm processing. However, this cut-off was defined by the SiD™ developers to guarantee a sufficient number of spermatozoa in the ICSI drop, enabling the software to evaluate and discriminate the population accurately and select the most suitable cells for injection. At the same time, this relatively low threshold broadens the scope of the study by allowing the inclusion of more heterogeneous samples, rather than limiting the analysis to those with high sperm concentrations.

On the other hand, this study has several important strengths. It is, to our knowledge, the first study to retrospectively assess both gametes using different AI tools while simultaneously evaluating subsequent embryo quality with multiple AI algorithms. Moreover, this preliminary analysis provides valuable insights, such as the observation that sperm contribution is more reliably studied in patients using their own oocytes, where oocyte quality is generally lower than in egg donation cycles. In aged oocytes, DNA repair mechanisms may be diminished, making differences between sperm of different quality more readily detectable [36]. Therefore, AI-assisted sperm selection could be particularly valuable in populations with poor oocyte quality, typically represented by advanced-age patients undergoing reproductive cycles with their own oocytes. However, this idea must be regarded strictly as speculative and hypothesis-generating, since our study was not designed or powered to directly address this question.

One of the main limitations of incorporating new AI-based technologies for sperm selection into routine practice is the need for specific training of embryologists. However, in the case of SiD™, the system is designed to be user-friendly and does not require extensive training [26]. Importantly, the ICSI procedure itself is performed in the conventional manner, with slight modifications to the established micromanipulation steps. Thus, the use of SiD™ does not complicate or alter the traditional ICSI procedure, but rather complements it by providing an additional layer of information that can be easily adopted in daily laboratory practice.

Overall, the potential to conduct real-time analysis of individual spermatozoa through a standardized automated tool, presents an opportunity to aid embryologists in selecting the optimal spermatozoa to microinject into the oocytes during the ICSI procedure, reducing

inter-user variability. Additionally, the biological patterns observed in our study are intriguing, but stronger and more definitive conclusions will require well-designed randomized controlled trials with adequate statistical power.

Abbreviations

AI	Artificial Intelligence
AISS	Artificial Intelligence Sperm Selection
ALH	Amplitude of Lateral Head Displacement
ART	Assisted Reproductive Treatments
ASEBIR	Association for the Study of Reproductive Biology
BMI	Body Mass Index
D5	Day 5
D6	Day 6
EIM	European IVF Monitoring Consortium
ESHRE	European Society of Human Reproduction and Embryology
FS	Fresh sample
ICM	Inner Cell Mass
ICSI	IntraCytoplasmic Sperm Injection
IMSI	Intracytoplasmic Morphologically Selected Sperm Injection
IVF	In Vitro Fertilization
GnRH	Gonadotropine Releaser Hormone
GM-sperm	Good-motility sperm
hCG	Human Chorionic Gonadotropin
HMG	Human Menopausal Gonadotropin
HMP	Head Movement Pattern
LH	Luteinizing Hormone
LIN	Linearity
PB	Polar Body
PGT-A	Preimplantation Genetic Test of Aneuploidies
PICSI	Physiological IntraCytoplasmic Sperm Injection
PM-sperm	Poor-motility sperm
PN	Pronuclei
PVP	Polyvinylpyrrolidone
SID™	Sperm ID
SS	Selected sample
STR	Straightness
TE	Trophectoderm
VSL	Straight-Line Velocity
VAP	Average Path Velocity
VCL	Curvilinear Velocity
WHO	World Health Organisation

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12958-025-01479-9>.

Supplementary Material 1
Supplementary Material 2
Supplementary Material 3
Supplementary Material 4
Supplementary Material 5
Supplementary Material 6
Supplementary Material 7

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Authors' contributions

LCS: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing.

TV: Investigation, Data curation, Validation, Supervision. EM: Investigation, Software, Validation, Supervision. TC: Investigation, Data curation, Writing – review & editing. JR: Visualization, Supervision, Project administration, Funding acquisition. MM: Conceptualization, Methodology, Validation, Resources, Data curation, Writing – review & editing, Visualization, Supervision, Project administration, Funding acquisition. All authors reviewed the manuscript.

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Data availability

The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

The procedure and protocol were approved by our institutional review board and ethical committee (code: 2105-VLC-037-MM) regulating and approving database analysis and clinical in vitro fertilization (IVF) procedures for research at IVIRMA Valencia. This project complies with Spanish law governing assisted reproductive technologies (14/2006).

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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